EA ELITE IAL MATHEMATICS

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WMA11/01 January 2024 Worked Solutions

Jan 2024 paper rebuilt with the worked solution after each question.

10
paper questions

WMA11
Pure 1

Single-paper solutions
standalone IAL route

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P1

PAST PAPER

WMA11/01 January 2024

Jan 2024 | 10 questions | 75 marks

10 questions	**75** marks	**Answers** worked solution after each question

Question 1

5 marks

Integration

1. Find

$$\int (2x - 5)(3x + 2)(2x + 5)\,\mathrm{d}x$$

writing your answer in simplest form.

(5)

(Total for Question 1 is 5 marks)

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Worked Solution - Question 1

Integration

Topic group

1. Expand two brackets first

$(2x-5)(3x+2)=6x^2-11x-10.$

2. Multiply by the third bracket

$(6x^2-11x-10)(2x+5)=12x^3+8x^2-75x-50.$

3. Integrate term by term

$$\int(12x^3+8x^2-75x-50)\,dx=3x^4+\frac{8}{3}x^3-\frac{75}{2}x^2-50x+c.$$

Final answer

$$3x^4+\frac{8}{3}x^3-\frac{75}{2}x^2-50x+c.$$

Question 2

5 marks

Basic Trigonometry

2. The triangle ABC is such that

- $AB = 15\,\text{cm}$
- $AC = 25\,\text{cm}$
- angle $BAC = \theta°$
- area triangle $ABC = 100\,\text{cm}^2$

(a) Find the value of $\sin \theta°$

(2)

Given that $\theta > 90$

(b) find the length of BC, in cm, to 3 significant figures.

(3)

(Total for Question 2 is 5 marks)

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Worked Solution - Question 2

Basic Trigonometry

Topic group

1. Use the triangle area formula

$100 = \frac{1}{2}(15)(25)\sin\theta.$

2. Find sine theta

$\sin\theta = \frac{200}{375} = \frac{8}{15}.$

3. Use theta greater than 90 degrees

Since $\theta > 90^\circ$, $\cos\theta$ is negative. So $\cos\theta = -\sqrt{1-\left(\frac{8}{15}\right)^2} = -\frac{\sqrt{161}}{15}.$

4. Apply the cosine rule

$BC^2 = 15^2 + 25^2 - 2(15)(25)\cos\theta = 850 + 50\sqrt{161}.$

5. Round the length

$BC = \sqrt{850 + 50\sqrt{161}} \approx 38.5\ \mathrm{cm}$ to 3 significant figures.

Final answer

$(a)\ \ \sin\theta = \frac{8}{15}.$

$(b)\ BC = 38.5\ \mathrm{cm}.$

Question 3

7 marks

Differentiation

3. The curve C has equation

$$y = \frac{5x^3 - 8}{2x^2} \qquad x > 0$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ writing your answer in simplest form.

(4)

The point $P(2, 4)$ lies on C.

(b) Find an equation for the tangent to C at P writing your answer in the form $ax + by + c = 0$, where a, b and c are integers.

(3)

(Total for Question 3 is 7 marks)

Worked Solution - Question 3

Differentiation

Topic group

1. Rewrite y

$$y = \frac{5x^3 - 8}{2x^2} = \frac{5}{2}x - 4x^{-2}.$$

2. Differentiate

$$\frac{dy}{dx} = \frac{5}{2} + 8x^{-3} = \frac{5}{2} + \frac{8}{x^3}.$$

3. Find the gradient at P

At $x = 2$, $\frac{dy}{dx} = \frac{5}{2} + \frac{8}{8} = \frac{7}{2}$.

4. Use point-gradient form

The tangent through $P(2, 4)$ is $y - 4 = \frac{7}{2}(x - 2)$.

5. Rearrange

$2y - 8 = 7x - 14$, so $7x - 2y - 6 = 0$.

Final answer

$(a)\ \frac{dy}{dx} = \frac{5}{2} + \frac{8}{x^3}$.

$(b)\ 7x - 2y - 6 = 0$.

Question 4

6 marks

Indices & Surds

4. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 2^x$, show that the equation

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4$$

can be written in the form

$$4p^2 - 33p + 8 = 0$$

(3)

(b) Hence solve

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4$$

(3)

(Total for Question 4 is 6 marks)

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Worked Solution - Question 4

Indices & Surds

Topic group

1. Substitute p equals 2 to the x

If $p = 2^x$, then $4^x = p^2$, $2^{x+3} = 8p$, and $2^{x-1} = \frac{p}{2}$.

2. Rewrite the equation

$2p^2 - 8p = 17\left(\frac{p}{2}\right) - 4$.

3. Clear the fraction

Multiplying by 2 gives $4p^2 - 16p = 17p - 8$, so $4p^2 - 33p + 8 = 0$.

4. Solve the quadratic

$4p^2 - 33p + 8 = (4p - 1)(p - 8)$, so $p = \frac{1}{4}$ or $p = 8$.

5. Return to x

$2^x = \frac{1}{4} = 2^{-2}$ gives $x = -2$, and $2^x = 8 = 2^3$ gives $x = 3$.

Final answer

$x = 3$ or $x = -2$.

Question 5

5.

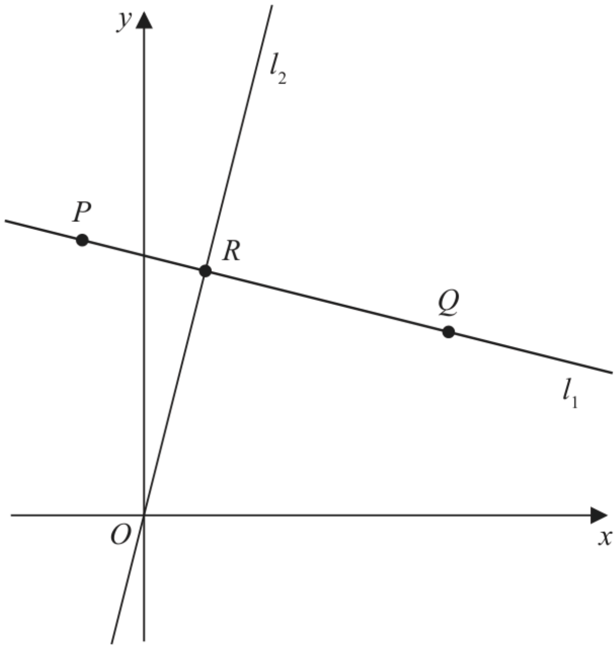

Figure 1

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The straight line l_1, shown in Figure 1, passes through the points $P(-2, 9)$ and $Q(10, 6)$.

(a) Find the equation of l_1, giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The straight line l_2 passes through the origin O and is perpendicular to l_1

The lines l_1 and l_2 meet at the point R as shown in Figure 1.

(b) Find the coordinates of R

(4)

(c) Find the exact area of triangle OPQ.

(3)

(Total for Question 5 is 10 marks)

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Worked Solution - Question 5

Straight Line

Topic group

1. Find the gradient of l1

$$m = \frac{6-9}{10-(-2)} = -\frac{3}{12} = -\frac{1}{4}.$$

2. Find the equation of l1

Using $P(-2, 9)$, $y = -\frac{1}{4}x + c$. Then $9 = \frac{1}{2} + c$, so $c = \frac{17}{2}$.

3. Find the equation of l2

A line perpendicular to gradient $-\frac{1}{4}$ has gradient 4. Since l_2 passes through the origin, l_2 is $y = 4x$.

4. Find R

At the intersection, $4x = -\frac{1}{4}x + \frac{17}{2}$. Hence $\frac{17}{4}x = \frac{17}{2}$, so $x = 2$ and $y = 8$.

5. Use coordinate area

For $O(0, 0)$, $P(-2, 9)$ and $Q(10, 6)$, the area is $\frac{1}{2}|(-2)(6) - 9(10)| = 51$.

Final answer

(a) $y = -\frac{1}{4}x + \frac{17}{2}$.

(b) $R = (2, 8)$.

(c) $\text{Area} = 51$.

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Question 6

6 marks

Trigonometric Functions

6.

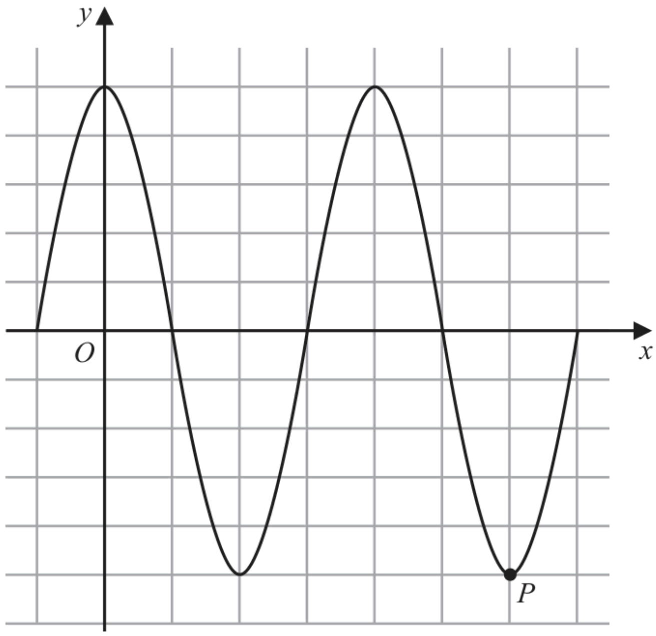

Figure 2

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5\cos x$$

with x being measured in degrees.

The point P, shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5\cos x - 2$

(ii) C_2 has equation $y = -5\cos x$

(4)

(Total for Question 6 is 6 marks)

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Worked Solution - Question 6

Trigonometric Functions

Topic group

1. Use the cosine graph

For $y = 5\cos x$, the minimum value is -5 and it occurs at $x = 180° + 360°n$.

2. Read P from the displayed minimum

The plotted point P is the displayed minimum at $x = 540°$, so $P = (540°, -5)$.

3. Translate the graph down

For $y = 5\cos x - 2$, the maximum value is $5 - 2 = 3$. The smallest positive x-coordinate for a maximum shown is $x = 360°$, so $Q = (360°, 3)$.

4. Reflect in the x-axis

For $y = -5\cos x$, a maximum occurs where $\cos x = -1$, first at $x = 180°$. The maximum value is 5, so $Q = (180°, 5)$.

Final answer

$(a)\ P = (540°, -5)$.

$(b)(i)\ Q = (360°, 3)$.

$(b)(ii)\ Q = (180°, 5)$.

Question 7

9 marks

Graphs of Functions

7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

(b) find the range of values of k.

(5)

(Total for Question 7 is 9 marks)

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Worked Solution - Question 7

Graphs of Functions

Topic group

1. Identify the asymptote

For $y = \dfrac{4}{x-k}$, the vertical asymptote is where the denominator is zero, so $x = k$
.

2. Find axis intercepts

At $x = 0$, $y = -\dfrac{4}{k}$, so the y-intercept is $(0, -\dfrac{4}{k})$. The graph has no x-intercept because the numerator is never zero.

3. Set up intersection with the line

If the line $y = 9 - x$ meets the curve, then $9 - x = \dfrac{4}{x-k}$.

4. Form a quadratic

$(9-x)(x-k) = 4$, which simplifies to $x^2 - (9+k)x + 9k + 4 = 0$.

5. Use no real intersection

For the line not to cross or touch the curve, the discriminant must be negative.

6. Solve the inequality

$(9+k)^2 - 4(9k+4) < 0$, so $k^2 - 18k + 65 < 0$. Since $k^2 - 18k + 65 = (k-5)(k-13)$, the range is $5

Final answer

(a) y-intercept $(0, -\dfrac{4}{k})$, no x-intercept, vertical asymptote $x = k$.

$(b)\ 5

Question 8

9 marks

Radians

8.

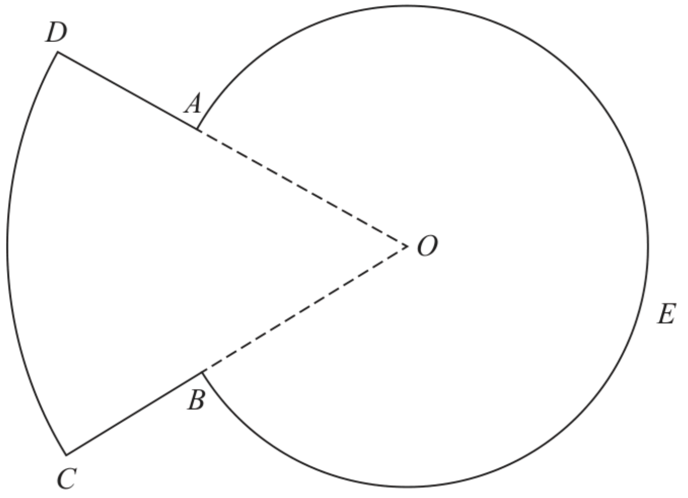

Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O.

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9\,\text{m}$
- $DA : AO = 3 : 5$

(a) show that $AO = 7.03\,\text{m}$ to 3 significant figures.

(3)

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

(Total for Question 8 is 9 marks)

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Worked Solution - Question 8

Radians

Topic group

1. Find the outer radius

For sector DOC, arc length $CD = 9$ and angle 0.8 radians, so $OD = \dfrac{9}{0.8} = 11.25\text{ m}$.

2. Use the ratio DA to AO

$DA : AO = 3 : 5$, so $OD = DA + AO$ represents 8 parts. Hence $AO = \dfrac{5}{8}(11.25) = 7.03125\text{ m} \approx 7.03\text{ m}$.

3. Find the straight side lengths

$DA = CB = OD - AO = 11.25 - 7.03125 = 4.21875\text{ m}$.

4. Find the major arc

The major sector $AOBEA$ has angle $2\pi - 0.8$, so arc $AEB = 7.03125(2\pi - 0.8)$.

5. Calculate the perimeter

$\text{Perimeter} = 9 + 2(4.21875) + 7.03125(2\pi - 0.8) \approx 56.0\text{ m}$.

6. Calculate the total area

$\text{Area} = \dfrac{1}{2}(11.25)^2(0.8) + \dfrac{1}{2}(7.03125)^2(2\pi - 0.8) \approx 186\text{ m}^2$ to the nearest whole number.

Final answer

(a) $AO = 7.03$ m.

(b) 56.0 m.

(c) $186\ \mathrm{m}^2$.

Question 9

10 marks

Quadratics

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at $(2, 20)$
- C_1 passes through the origin

(a) sketch a graph of C_1 showing the coordinates of any points where C_1 cuts the coordinate axes,

(2)

(b) find an expression for $f(x)$.

(3)

The curve C_2 has equation $y = x(x^2 - 4)$

Curve C_1 and C_2 meet at the origin, and at the points P and Q

Given that the x coordinate of the point P is negative,

(c) using algebra and showing all stages of your working, find the coordinates of P

(5)

(Total for Question 9 is 10 marks)

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Worked Solution - Question 9

Quadratics

Topic group

1. Use symmetry for the sketch

The maximum is at $(2, 20)$ and the curve passes through $(0, 0)$. Since the axis of symmetry is $x = 2$, the other x-intercept is $(4, 0)$.

2. Use completed-square form

Let $f(x) = a(x-2)^2 + 20$.

3. Use the origin

$0 = a(0-2)^2 + 20$, so $4a = -20$ and $a = -5$.

4. Write f(x)

$f(x) = -5(x-2)^2 + 20 = -5x^2 + 20x = -5x(x-4)$.

5. Equate the two curves

$-5x^2 + 20x = x(x^2-4) = x^3 - 4x$.

6. Factorise

$x^3 + 5x^2 - 24x = 0$, so $x(x^2 + 5x - 24) = 0$ and $x(x+8)(x-3) = 0$.

7. Use the negative x-coordinate

The negative x-coordinate is $x = -8$.

8. Find y

$y = x(x^2-4) = (-8)(64-4) = -480$, so $P = (-8, -480)$.

Final answer

(a) downward parabola through $(0, 0)$ and $(4, 0)$ with maximum $(2, 20)$.

(b) $f(x) = -5x(x - 4)$.

(c) $P = (-8, -480)$.

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Question 10

8 marks

Integration

10. **In this question you must show all stages of your working.**

The curve C has equation $y = f(x), x > 0$

Given that

- the point $P\left(2, 8\sqrt{2}\right)$ lies on C
- $f'(x) = 4\sqrt{x^3} + \dfrac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k,

(4)

(b) find $f(x)$, giving your answer in simplest form.

(4)

(Total for Question 10 is 8 marks)

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Worked Solution - Question 10

Integration

Topic group

1. Differentiate f prime

$f'(x) = 4x^{3/2} + kx^{-2}$, so $f''(x) = 6x^{1/2} - 2kx^{-3}$.

2. Use f double prime equals zero at P

At P, $x = 2$ and $f''(2) = 0$, so $6\sqrt{2} - 2k(2^{-3}) = 0$.

3. Find k

$6\sqrt{2} - \dfrac{k}{4} = 0$, hence $k = 24\sqrt{2}$.

4. Integrate f prime

$f(x) = \int \left(4x^{3/2} + 24\sqrt{2}x^{-2}\right) dx = \dfrac{8}{5}x^{5/2} - 24\sqrt{2}x^{-1} + c$.

5. Use the point P

$P(2, 8\sqrt{2})$ lies on C, so $8\sqrt{2} = \dfrac{8}{5}(2^{5/2}) - \dfrac{24\sqrt{2}}{2} + c$.

6. Find c

$8\sqrt{2} = \dfrac{32\sqrt{2}}{5} - 12\sqrt{2} + c$, so $c = \dfrac{68\sqrt{2}}{5}$.

Final answer

$(a)\ k = 24\sqrt{2}$.

$(b)\ f(x) = \dfrac{8}{5}x^{5/2} - \dfrac{24\sqrt{2}}{x} + \dfrac{68\sqrt{2}}{5}$.